

how does eq. 2 follow from eq. 1?

$$\text{eq. 1 } 8x^2 = (2n+1)^2 - 1$$

$$\text{eq. 2 } (2n+1)/x = 8^{.5} + 1/(x^2(8^{.5} + (2n+1)/x))$$

Start from **eq. 1**

$$8x^2 = (2n + 1)^2 - 1.$$

Step 1: Factor the right-hand side

$$(2n + 1)^2 - 1 = ((2n + 1) - 1)((2n + 1) + 1) = (2n)(2n + 2).$$

But for the next step it's more useful to keep it as

$$(2n + 1)^2 - 1.$$

Rewrite eq. 1 as

$$(2n + 1)^2 = 8x^2 + 1.$$

Step 2: Divide by x^2

$$\left(\frac{2n + 1}{x}\right)^2 = 8 + \frac{1}{x^2}.$$

Step 3: Take square roots

$$\frac{2n + 1}{x} = \sqrt{8 + \frac{1}{x^2}}.$$

Step 4: Rationalize the expression

Write

$$\sqrt{8 + \frac{1}{x^2}} = \sqrt{8} \sqrt{1 + \frac{1}{8x^2}} = \sqrt{8} \left(1 + \frac{1}{\sqrt{8}x^2(\sqrt{8} + \sqrt{8 + \frac{1}{x^2}})} \right).$$

Equivalently, using the identity

$$\sqrt{A+B} - \sqrt{A} = \frac{B}{\sqrt{A+B} + \sqrt{A}},$$

with $A = 8$ and $B = \frac{1}{x^2}$, we get

$$\sqrt{8 + \frac{1}{x^2}} = \sqrt{8} + \frac{1}{x^2 \left(\sqrt{8} + \sqrt{8 + \frac{1}{x^2}} \right)}.$$

Step 5: Substitute back

Since $\sqrt{8 + \frac{1}{x^2}} = \frac{2n+1}{x}$,

$$\frac{2n+1}{x} = \sqrt{8} + \frac{1}{x^2 \left(\sqrt{8} + \frac{2n+1}{x} \right)}.$$

This is exactly **eq. 2**.

